

Radial Extremality for LRU Caching and the Fill–Holst Conjecture

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Abstract

For the independent reference model with popularity vector $p \in \Delta_N^\circ$, let $H_C(p)$ denote the exact stationary hit rate of an LRU cache of capacity C . We prove that, for every $1 \leq C < N$, the uniform popularity vector is the unique global minimizer of H_C on the interior simplex. More sharply, along every nonconstant segment from the uniform vector to an interior point, the LRU hit rate is strictly increasing. The proof uses the standard exponential-age representation of the stationary LRU cache and gives an explicit positive pair-square formula for the radial derivative. Equivalently, for the move-to-front rule, the stationary search-cost distribution improves strictly in the usual stochastic order along every nonconstant ray away from uniform. This proves the radial restriction of the Fill–Holst Schur-concavity conjecture for move-to-front search-cost tails. In particular, all LRU miss probabilities and all nonconstant nondecreasing stack-depth costs decrease strictly along such rays. The result is radial rather than Schur-convex: full majorization monotonicity for LRU is known to fail, and the proof identifies the special positivity that survives on rays from the uniform vector.

1 Introduction

The independent reference model (IRM) is the classical model in which successive requests are independent and item i is requested with probability p_i in discrete time, equivalently with rate p_i in the rate-one Poissonized model. Under LRU, the stationary stack order has an exact exponential-age representation: if $A_1, \dots, A_N$ are independent exponential random variables with $A_i \sim \text{Exp}(p_i)$, then the stationary cache of capacity C consists of the C items with smallest ages. This representation follows by Poissonizing the request sequence into independent Poisson streams of rates p_i and using their backward recurrence times; conditioning on the next arrival recovers the original discrete request distribution. The hit rate is therefore an exact finite-dimensional function of the popularity vector, not a characteristic-time or Che-type approximation. Exact formulae and asymptotic approximations for LRU under the IRM go back to King, Fagin, Fagin–Price, and Flajolet–Gardy–Thimonier [9, 3, 4, 6]; Che–Tung–Wang introduced a now-standard characteristic-time approximation for hierarchical Web caching [2], Jelenković studied asymptotic MTF/LRU fault probabilities for broad popularity classes [8], and Berthet discusses the identity of the King and Flajolet–Gardy–Thimonier formulae for exact LRU miss-rate computation [1].

A natural monotonicity question asks whether making the popularity vector more uneven must improve the LRU hit rate. In the language of majorization [11], this would assert Schur-convexity of the hit rate, or equivalently Schur-concavity of the miss rate. A closely related move-to-front literature considered the same distributional question in search-cost language. Under independent requests, the stationary move-to-front list is the LRU recency order, and the LRU miss probability for a cache of capacity C is the tail probability that the move-to-front search cost exceeds C . Fill and Holst studied the stationary move-to-front search-cost distribution and its cache-fault interpretation,

and Hildebrand later addressed their conjecture that each search-cost tail is Schur-concave in the popularity vector [5, 7]. Hildebrand disproved that conjecture.

Thus full Schur-concavity is too strong even for the natural MTF/LRU tail probabilities. Vanichpun and Makowski also showed, in a broader caching setting, that majorization monotonicity may fail under LRU [12, 10]. The theorem below identifies a sharp positive remnant of that false global principle: although arbitrary majorization directions can move the LRU hit rate in the wrong direction, every straight-line move away from the uniform popularity vector strictly increases the hit rate. Equivalently, the radial restriction of the Fill–Holst tail-concavity conjecture is true for MTF/LRU.

The proof is deliberately finite and exact. We first derive a residual subset expansion for $H_C(p)$. A variance identity on each residual subset converts that expansion into a pair-square decomposition around the uniform vector. Radial differentiation then produces an explicit pair-square formula

$$\frac{d}{d\theta} H_C(u + \theta(q - u)) = \frac{1}{N\theta} \sum_{1 \leq a < b \leq N} (p_a(\theta) - p_b(\theta))^2 K_{ab}^{(C)}(p(\theta)),$$

where every kernel $K_{ab}^{(C)}$ is strictly positive. The positivity of the kernels is proved by converting the alternating subset sums into positive product integrals. This gives a local certificate for radial monotonicity, not merely a global comparison.

After proving the main theorem, we record its distributional consequences for the full LRU stack-depth distribution, or equivalently for the stationary move-to-front search-cost distribution. We also include an appendix with an independent occupancy-Jacobian proof. That proof explains at the differential level why radial monotonicity is compatible with failure of Schur-convexity: the asymmetric occupancy Jacobian is not positive in every tangent direction, but its sign-indefinite minor collapses to a positive square along rays from the uniform vector.

2 Setup and main results

Let

$$\Delta_N^\circ = \left\{ p = (p_1, \dots, p_N) : p_i > 0, \sum_{i=1}^N p_i = 1 \right\}, \quad u = \left(\frac{1}{N}, \dots, \frac{1}{N} \right).$$

For $R \subseteq [N] := \{1, \dots, N\}$, write

$$p_R = \sum_{i \in R} p_i, \quad p_\emptyset = 0.$$

Let $A_1, \dots, A_N$ be independent exponential random variables with $A_i \sim \text{Exp}(p_i)$, and let S_C be the set of the C indices with smallest ages. The exact stationary LRU hit rate is

$$H_C(p) = \mathbb{E} \left[\sum_{i \in S_C} p_i \right]. \tag{1}$$

Equivalently, since the next request has distribution p independently of the stationary stack, $H_C(p)$ is the stationary probability that the next requested item is already in an LRU cache of capacity C .

Theorem 2.1 (Radial LRU extremality). *Let $N \geq 2$, let $1 \leq C < N$, and let $q \in \Delta_N^\circ$ be nonuniform. Define*

$$p(\theta) = u + \theta(q - u), \quad 0 \leq \theta \leq 1.$$

Then

$$\frac{d}{d\theta} H_C(p(\theta)) > 0, \quad 0 < \theta \leq 1.$$

Consequently $\theta \mapsto H_C(p(\theta))$ is strictly increasing on $(0, 1]$.

Corollary 2.2 (Uniform popularity is the unique worst case). *Let $N \geq 2$ and $1 \leq C < N$. For every nonuniform $p \in \Delta_N^\circ$,*

$$H_C(p) > H_C(u) = \frac{C}{N}.$$

Thus u is the unique global minimizer of the LRU hit rate on Δ_N° . Equivalently, the LRU miss rate $1 - H_C(p)$ is uniquely maximized at u on Δ_N° .

Proof. Apply Theorem 2.1 with $q = p$. Then $p(0) = u$ and $p(1) = p$. By Lemma 3.1, H_C is continuous along this segment, and Theorem 2.1 gives strict monotonicity on $(0, 1]$; hence $H_C(p) > H_C(u)$. Under u , symmetry gives $\mathbb{P}(i \in S_C) = C/N$ for every item i , and therefore

$$H_C(u) = \sum_{i=1}^N \frac{1}{N} \cdot \frac{C}{N} = \frac{C}{N}.$$

□

The proof of Theorem 2.1 proceeds through a stronger positive kernel formula. Put

$$L = N - C + 1.$$

Thus $2 \leq L \leq N$. For $a < b$ and $p \in \Delta_N^\circ$, define

$$K_{ab}^{(C)}(p) = N \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} (-1)^{|R|-L} \binom{|R|-2}{|R|-L} \left(\frac{1}{|R| p_R} + \frac{1}{N p_R^2} \right). \quad (2)$$

Theorem 2.3 (Positive pair-kernel derivative formula). *For every $p \in \Delta_N^\circ$ and every $a < b$,*

$$K_{ab}^{(C)}(p) > 0.$$

Moreover, if $p(\theta) = u + \theta(q - u)$ with $0 < \theta \leq 1$, then

$$\frac{d}{d\theta} H_C(p(\theta)) = \frac{1}{N\theta} \sum_{1 \leq a < b \leq N} (p_a(\theta) - p_b(\theta))^2 K_{ab}^{(C)}(p(\theta)). \quad (3)$$

Theorem 2.1 follows immediately from Theorem 2.3: if $q \neq u$ and $0 < \theta \leq 1$, then $p(\theta)$ is nonuniform, so at least one pair difference in (3) is nonzero.

3 A residual-subset formula for the hit rate

We first convert the exponential-clock representation into a finite alternating subset expansion. This is the only place where the cache capacity enters in an essential way.

Lemma 3.1 (Residual-subset expansion). *Let $1 \leq C < N$ and $L = N - C + 1$. For $L \leq m \leq N$, set*

$$\alpha_m = (-1)^{m-L} \binom{m-2}{m-L}.$$

Then, for every $p \in \Delta_N^\circ$,

$$H_C(p) = \sum_{m=L}^N \alpha_m \sum_{\substack{R \subseteq [N] \\ |R|=m}} \frac{\sum_{i \in R} p_i^2}{p_R}. \quad (4)$$

Proof. Fix an item i . Its contribution to the hit rate is $p_i \mathbb{P}(i \in S_C)$. Conditioning on $A_i = t$ gives

$$p_i \mathbb{P}(i \in S_C) = p_i^2 \int_0^\infty e^{-p_i t} \sum_{\substack{S \subseteq [N] \setminus \{i\} \\ |S| \leq C-1}} \prod_{j \in S} (1 - e^{-p_j t}) \prod_{\ell \notin S \cup \{i\}} e^{-p_\ell t} dt. \quad (5)$$

Here S is the set of other items whose ages are smaller than t ; item i is in the cache precisely when $|S| \leq C - 1$.

Expand

$$\prod_{j \in S} (1 - e^{-p_j t}) = \sum_{A \subseteq S} (-1)^{|A|} e^{-p_A t}.$$

For a term indexed by $A \subseteq S$, put $B = S \setminus A$. The total exponent in (5) is

$$p_i + p_A + p_{[N] \setminus (S \cup \{i\})} = 1 - p_B.$$

Now fix $B \subseteq [N] \setminus \{i\}$ with $b = |B| \leq C - 1$. The set A may be any subset of $[N] \setminus (\{i\} \cup B)$ with $|A| \leq C - 1 - b$. Hence the coefficient of $e^{-(1-p_B)t}$ is

$$\sum_{h=0}^{C-1-b} (-1)^h \binom{N-1-b}{h}.$$

Since $C < N$, the elementary identity

$$\sum_{h=0}^q (-1)^h \binom{M}{h} = (-1)^q \binom{M-1}{q}, \quad 0 \leq q < M,$$

shows that this coefficient is

$$(-1)^{C-1-b} \binom{N-2-b}{C-1-b}.$$

Therefore

$$p_i \mathbb{P}(i \in S_C) = p_i^2 \sum_{\substack{B \subseteq [N] \setminus \{i\} \\ |B| \leq C-1}} (-1)^{C-1-|B|} \binom{N-2-|B|}{C-1-|B|} \frac{1}{1-p_B}.$$

Set $R = [N] \setminus B$. Then $i \in R$, $p_R = 1 - p_B$, and $m := |R| = N - |B|$ satisfies $m \geq N - C + 1 = L$. Moreover

$$(-1)^{C-1-|B|} \binom{N-2-|B|}{C-1-|B|} = (-1)^{m-L} \binom{m-2}{m-L} = \alpha_m.$$

Summing over i and grouping by R gives (4). $\square$

4 Pair-square decomposition

The residual expansion separates cleanly into the uniform contribution and pairwise fluctuations.

Lemma 4.1 (Pair-square expansion of H_C). *For every $p \in \Delta_N^\circ$,*

$$H_C(p) = \frac{C}{N} + \sum_{1 \leq a < b \leq N} (p_a - p_b)^2 J_{ab}^{(C)}(p), \quad (6)$$

where

$$J_{ab}^{(C)}(p) = \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} (-1)^{|R|-L} \binom{|R|-2}{|R|-L} \frac{1}{|R| p_R}. \quad (7)$$

Proof. For any nonempty finite set R with $m = |R|$,

$$\sum_{i \in R} p_i^2 = \frac{p_R^2}{m} + \frac{1}{m} \sum_{\substack{a < b \\ a, b \in R}} (p_a - p_b)^2. \quad (8)$$

This follows from

$$\sum_{\substack{a < b \\ a, b \in R}} (p_a - p_b)^2 = m \sum_{i \in R} p_i^2 - p_R^2.$$

Substituting (8) into (4) gives

$$\begin{aligned} H_C(p) &= \sum_{m=L}^N \alpha_m \sum_{\substack{R \subseteq [N] \\ |R|=m}} \frac{p_R}{m} \\ &\quad + \sum_{1 \leq a < b \leq N} (p_a - p_b)^2 \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} \frac{\alpha_{|R|}}{|R| p_R}. \end{aligned}$$

The second term is the pair-square term in (6). The first term is constant on the simplex, because

$$\sum_{\substack{R \subseteq [N] \\ |R|=m}} p_R = \binom{N-1}{m-1} \sum_{i=1}^N p_i = \binom{N-1}{m-1}.$$

To identify the constant, evaluate at $p = u$. By symmetry, $H_C(u) = C/N$, and at $p = u$ all pair differences vanish. Hence the constant is C/N . $\square$

5 Radial differentiation

Let $q \in \Delta_N^\circ$ and set

$$p(\theta) = u + \theta(q - u), \quad 0 < \theta \leq 1.$$

For readability, write $x = p(\theta)$ during the differentiation. If $R \subseteq [N]$ and $m = |R|$, then

$$\theta \frac{dx_i}{d\theta} = x_i - \frac{1}{N}, \quad \theta \frac{dx_R}{d\theta} = x_R - \frac{m}{N}, \quad \theta \frac{d}{d\theta}(x_a - x_b) = x_a - x_b. \quad (9)$$

Consequently,

$$\begin{aligned}\theta \frac{d}{d\theta} \left(\frac{(x_a - x_b)^2}{mx_R} \right) &= (x_a - x_b)^2 \left(\frac{2}{mx_R} - \frac{x_R - m/N}{mx_R^2} \right) \\ &= (x_a - x_b)^2 \left(\frac{1}{mx_R} + \frac{1}{Nx_R^2} \right).\end{aligned}\tag{10}$$

Differentiating (6) term by term and using (10) gives

$$\frac{d}{d\theta} H_C(p(\theta)) = \frac{1}{\theta} \sum_{1 \leq a < b \leq N} (x_a - x_b)^2 \sum_{\substack{R \subseteq [N]: \{a,b\} \subseteq R \\ |R| \geq L}} (-1)^{|R|-L} \binom{|R|-2}{|R|-L} \left(\frac{1}{|R|x_R} + \frac{1}{Nx_R^2} \right).$$

This is precisely (3), with $K_{ab}^{(C)}$ defined by (2). It remains only to prove that these kernels are strictly positive.

6 Strict positivity of the pair kernels

Fix $p \in \Delta_N^\circ$ and a pair $a < b$. Let

$$T = [N] \setminus \{a, b\}, \quad s = p_a + p_b, \quad r = L - 2 = N - C - 1.$$

Since $1 \leq C < N$, we have $0 \leq r \leq N - 2 = |T|$. Writing $R = \{a, b\} \cup U$ with $U \subseteq T$, the coefficient in (2) becomes

$$(-1)^{|R|-L} \binom{|R|-2}{|R|-L} = (-1)^{|U|-r} \binom{|U|}{r},$$

where the binomial coefficient is zero if $|U| < r$. Hence

$$\frac{1}{N} K_{ab}^{(C)}(p) = \Phi_r + \frac{1}{N} \Psi_r,\tag{11}$$

where

$$\Phi_r := \sum_{U \subseteq T} (-1)^{|U|-r} \binom{|U|}{r} \frac{1}{(|U|+2)(s+p_U)},\tag{12}$$

$$\Psi_r := \sum_{U \subseteq T} (-1)^{|U|-r} \binom{|U|}{r} \frac{1}{(s+p_U)^2}.\tag{13}$$

We prove that both Φ_r and Ψ_r are strictly positive.

For $t \geq 0$ and $0 \leq y \leq 1$, define

$$B_r(y, t) = \sum_{U \subseteq T} (-1)^{|U|-r} \binom{|U|}{r} y^{|U|} e^{-p_U t}.\tag{14}$$

Let

$$P_t(y) = \prod_{j \in T} (1 - ye^{-p_j t}).$$

Since

$$P_t(y) = \sum_{U \subseteq T} (-1)^{|U|} y^{|U|} e^{-p_U t},$$

we have

$$B_r(y, t) = \frac{(-1)^r y^r}{r!} P_t^{(r)}(y). \quad (15)$$

Differentiating the product gives the positive form

$$B_r(y, t) = y^r \sum_{\substack{W \subseteq T \\ |W|=r}} e^{-p_W t} \prod_{j \in T \setminus W} (1 - y e^{-p_j t}). \quad (16)$$

For $t > 0$ and $0 < y \leq 1$, every factor $1 - y e^{-p_j t}$ is strictly positive, because $p_j > 0$. Therefore

$$B_r(y, t) > 0, \quad t > 0, \quad 0 < y \leq 1. \quad (17)$$

Now use the elementary integral identities

$$\frac{1}{s + p_U} = \int_0^\infty e^{-(s+p_U)t} dt, \quad \frac{1}{|U| + 2} = \int_0^1 y^{|U|+1} dy,$$

and

$$\frac{1}{(s + p_U)^2} = \int_0^\infty t e^{-(s+p_U)t} dt.$$

Since all sums are finite, we may interchange sums and integrals. From (12) and (14),

$$\Phi_r = \int_0^\infty e^{-st} \int_0^1 y B_r(y, t) dy dt.$$

By (17), the integrand is strictly positive for $t > 0$ and $0 < y < 1$. Hence $\Phi_r > 0$. Similarly, from (13) and (14),

$$\Psi_r = \int_0^\infty t e^{-st} B_r(1, t) dt > 0.$$

Combining these inequalities with (11) gives $K_{ab}^{(C)}(p) > 0$. This proves Theorem 2.3, and hence Theorem 2.1.

Remark 6.1 (Endpoint capacities). For $C = 1$, one has $L = N$, and the residual formula reduces to

$$H_1(p) = \sum_{i=1}^N p_i^2.$$

For $C = N$, the hit rate is identically 1, so strict radial monotonicity is not a meaningful statement. The theorem therefore covers exactly the nontrivial range $1 \leq C < N$.

7 Move-to-front search costs and miss-probability consequences

The radial theorem has a useful distributional interpretation. Under independent requests, the stationary move-to-front list is the same recency ordering used by LRU, namely the ordering of all N items from most recently requested to least recently requested. Let $D = D(p)$ be the stationary move-to-front search cost, equivalently the LRU stack depth of the next requested item in stationarity. Then $D \in \{1, \dots, N\}$, and a cache of capacity C hits exactly when $D \leq C$. Therefore

$$\mathbb{P}_p(D \leq C) = H_C(p), \quad 1 \leq C \leq N. \quad (18)$$

Here $H_N \equiv 1$, while the nontrivial thresholds are $1 \leq C < N$.

Corollary 7.1 (Radial stochastic improvement for move-to-front search cost). *Let $N \geq 2$, let $q \in \Delta_N^\circ$ be nonuniform, and put $p(\theta) = u + \theta(q - u)$. If $0 < \theta_1 < \theta_2 \leq 1$, then the move-to-front search cost under $p(\theta_2)$ is strictly smaller than the move-to-front search cost under $p(\theta_1)$ in the usual stochastic order: for every $1 \leq C < N$,*

$$\mathbb{P}_{p(\theta_2)}(D \leq C) > \mathbb{P}_{p(\theta_1)}(D \leq C),$$

or equivalently

$$\mathbb{P}_{p(\theta_2)}(D > C) < \mathbb{P}_{p(\theta_1)}(D > C).$$

Proof. Identity (18) gives $\mathbb{P}_{p(\theta)}(D \leq C) = H_C(p(\theta))$. For $1 \leq C < N$, Theorem 2.1 shows that this quantity is strictly increasing in θ . Taking complements gives the tail form. $\square$

Corollary 7.2 (Nondecreasing search-cost functionals). *Under the hypotheses of Corollary 7.1, for every nondecreasing function $g : \{1, \dots, N\} \rightarrow \mathbb{R}$,*

$$\mathbb{E}_{p(\theta_2)}g(D) \leq \mathbb{E}_{p(\theta_1)}g(D).$$

The inequality is strict whenever g is not constant. In particular, the LRU miss probability

$$M_C(p) := 1 - H_C(p) = \mathbb{P}_p(D > C)$$

is strictly decreasing along every nonconstant ray away from uniform for each $1 \leq C < N$, and the expected move-to-front search cost

$$\mathbb{E}_p D = 1 + \sum_{c=1}^{N-1} M_c(p)$$

is strictly decreasing along every such ray.

Proof. Use the discrete tail-sum decomposition

$$\mathbb{E}g(D) = g(1) + \sum_{c=1}^{N-1} (g(c+1) - g(c))\mathbb{P}(D > c). \quad (19)$$

If g is nondecreasing, every coefficient in the sum is nonnegative, and Corollary 7.1 gives the asserted monotonicity. If g is not constant, at least one coefficient $g(c+1) - g(c)$ is positive, and the corresponding tail probability decreases strictly. The statements about miss probability and expected search depth are the special cases $g = \mathbf{1}_{\{C+1, \dots, N\}}$ and $g(d) = d$. $\square$

In this language, Theorem 2.1 proves the radial restriction of the Fill–Holst tail-concavity conjecture. Hildebrand’s counterexample shows that full Schur-concavity of the search-cost tails is false. The present result shows that the conjectured monotonicity nevertheless survives on every segment emanating from the uniform distribution.

8 Why the theorem is radial, not Schur-convex

The preceding result should not be interpreted as a disguised Schur-convexity theorem. Full Schur-convexity would require monotonicity under all elementary majorization transfers, not merely along rays from the uniform vector. The Fill–Holst tail conjecture is false [7], and the known Vanichpun–Makowski examples show that LRU can violate the corresponding majorization monotonicity [12, 10].

The point is that the radial direction from uniform aligns with a local asymmetry in the LRU occupancy map.

Appendix A gives the details, but the local mechanism is simple to state. Let $\pi_k(\lambda)$ denote the probability that item k is present in the stationary LRU cache when the exponential rates are λ . For $i \neq k$, the off-diagonal sensitivity has the form

$$\frac{\partial \pi_k}{\partial \lambda_i}(\lambda) = -\lambda_k G_{ik}(\lambda), \quad G_{ik}(\lambda) = G_{ki}(\lambda) > 0.$$

Thus the occupancy Jacobian is generally asymmetric. For an arbitrary tangent direction δ with $\sum_i \delta_i = 0$, the occupancy-sensitivity part of the directional derivative contains

$$\sum_{i < k} G_{ik}(\lambda)(\lambda_i - \lambda_k)(\lambda_k \delta_i - \lambda_i \delta_k),$$

and the minor $\lambda_k \delta_i - \lambda_i \delta_k$ has no fixed sign. Along the radial path $\lambda = u + \theta(q - u)$, however,

$$\delta_i = \frac{\lambda_i - 1/N}{\theta}, \quad \lambda_k \delta_i - \lambda_i \delta_k = \frac{\lambda_i - \lambda_k}{N\theta},$$

so the sign-indefinite term collapses to the positive square

$$\frac{1}{N\theta} \sum_{i < k} G_{ik}(\lambda)(\lambda_i - \lambda_k)^2.$$

This is the local reason radial monotonicity survives even though full Schur-convexity fails.

9 Further directions

The distinction between global Schur-concavity and radial monotonicity suggests a broader structural problem. For policies such as RANDOM replacement and FIFO, stronger Schur-concavity results are already known under the IRM, so radial monotonicity follows immediately [12, 10]. LRU/MTF is different: full Schur-concavity fails, but the present paper proves strict radial monotonicity. This raises the corresponding radial question for other self-organizing policies for which full Schur-concavity fails or remains unknown. Identifying the finite-state structure that controls radial monotonicity for such policies is left for future work.

A An occupancy-Jacobian proof

This appendix gives an independent proof of Theorem 2.1. The main proof above is shorter, but the Jacobian proof isolates the local mechanism behind the radial positivity.

Fix $\lambda \in \mathbb{R}_{>0}^N$, and let $A_i \sim \text{Exp}(\lambda_i)$ be independent. Let $S_C(\lambda)$ be the set of the C indices with smallest ages, and define

$$\pi_k(\lambda) = \mathbb{P}(k \in S_C(\lambda)).$$

The map π_k is homogeneous of degree zero, and

$$\sum_{k=1}^N \pi_k(\lambda) = C, \quad H_C(p) = \sum_{k=1}^N p_k \pi_k(p). \quad (20)$$

The first identity holds because $|S|_C = C$ deterministically. Differentiating it gives

$$\sum_{k=1}^N \frac{\partial \pi_k}{\partial \lambda_i}(\lambda) = 0 \quad \text{for every } i. \quad (21)$$

Lemma A.1 (Occupancy sensitivity). *For $i \neq k$ and any $\lambda \in \mathbb{R}_{>0}^N$,*

$$\frac{\partial \pi_k}{\partial \lambda_i}(\lambda) = -\lambda_k G_{ik}(\lambda), \quad (22)$$

where

$$G_{ik}(\lambda) = \int_0^\infty t e^{-(\lambda_i + \lambda_k)t} \mathbb{P}(M_{ik}(t) = C - 1) dt,$$

and

$$M_{ik}(t) = \#\{j \notin \{i, k\} : A_j < t\}.$$

The kernel is symmetric and nonnegative, $G_{ik} = G_{ki} \geq 0$. If $1 \leq C < N$, then $G_{ik} > 0$ for every pair $i \neq k$.

Proof. Conditioning on $A_k = t$ gives

$$\pi_k(\lambda) = \int_0^\infty \lambda_k e^{-\lambda_k t} \Phi_k(t; \lambda) dt,$$

where

$$\Phi_k(t; \lambda) = \mathbb{P} \left(\sum_{j \neq k} \mathbf{1}_{\{A_j < t\}} \leq C - 1 \right).$$

For $i \neq k$, write $M_{ik}(t) = \sum_{j \notin \{i, k\}} \mathbf{1}_{\{A_j < t\}}$. Splitting off the i th indicator and putting $x_i(t) = 1 - e^{-\lambda_i t}$,

$$\Phi_k = \mathbb{P}(M_{ik} \leq C - 1) - x_i(t) \mathbb{P}(M_{ik} = C - 1).$$

Hence

$$\frac{\partial \Phi_k}{\partial \lambda_i} = -t e^{-\lambda_i t} \mathbb{P}(M_{ik}(t) = C - 1).$$

Differentiating under the integral is justified by dominated convergence, for example on compact subsets of $\mathbb{R}_{>0}^N$: the differentiated integrand is bounded by an integrable multiple of $t e^{-\eta t}$ for some $\eta > 0$. This gives (22). The symmetry $G_{ik} = G_{ki}$ is immediate from the integral expression. For strict positivity, note that when $1 \leq C < N$, we have $0 \leq C - 1 \leq N - 2$. For every $t > 0$, each event probability in the Poisson-binomial mass $\mathbb{P}(M_{ik}(t) = C - 1)$ is a nonempty sum of strictly positive products, so the integrand is strictly positive on $(0, \infty)$. $\square$

The diagonal entries are determined by (21):

$$\frac{\partial \pi_i}{\partial \lambda_i}(\lambda) = \sum_{k \neq i} \lambda_k G_{ik}(\lambda). \quad (23)$$

We also need the elementary monotonicity of occupancy probabilities.

Lemma A.2 (Ordering law). *For every permutation $\sigma = (\sigma_1, \dots, \sigma_N)$ of $[N]$,*

$$\mathbb{P}(A_{\sigma_1} < A_{\sigma_2} < \dots < A_{\sigma_N}) = \prod_{r=1}^N \frac{\lambda_{\sigma_r}}{\sum_{m=r}^N \lambda_{\sigma_m}}.$$

Equivalently, the exponential-clock order is generated by repeatedly choosing the next index from the remaining indices with probability proportional to its rate.

Proof. The first clock to ring has label j with probability $\lambda_j / \sum_m \lambda_m$. Conditional on that first label, the residual waiting times of the remaining clocks are again independent exponentials with their original rates, by memorylessness. Iterating gives the displayed product. $\square$

Lemma A.3 (Occupancy monotonicity). *If $\lambda_k \geq \lambda_i$, then $\pi_k(\lambda) \geq \pi_i(\lambda)$. Consequently,*

$$(\lambda_a - \lambda_b)(\pi_a(\lambda) - \pi_b(\lambda)) \geq 0 \quad \text{for all } a, b.$$

Proof. It suffices to compare the cases in which exactly one of i, k belongs to the first C positions of the exponential-clock order, since the cases in which both are cached or both are not cached cancel in $\pi_k - \pi_i$. Let $\mathcal{A}$ be the set of orderings in which i appears in one of the first C positions and k appears after position C , and let $\mathcal{B}$ be the analogous set with i and k interchanged. Swapping the labels i and k gives a bijection $\mathcal{A} \rightarrow \mathcal{B}$.

Fix $\sigma \in \mathcal{A}$. Suppose i occurs at position $r \leq C$ and k occurs at position $s > C$. Let τ be obtained from σ by swapping the labels i and k , and put $d = \lambda_k - \lambda_i \geq 0$. In the product formula of Lemma A.2, the numerator factors involving i and k cancel in the ratio $\mathbb{P}(\tau)/\mathbb{P}(\sigma)$, and all denominator factors outside positions $r+1, \dots, s$ are identical. For $m = r+1, \dots, s$, the remaining-rate denominator under τ is smaller by exactly d than the corresponding denominator under σ . Hence

$$\frac{\mathbb{P}(\tau)}{\mathbb{P}(\sigma)} = \prod_{m=r+1}^s \frac{D_m}{D_m - d} \geq 1,$$

where D_m denotes the corresponding remaining-rate denominator under σ . Summing over the paired orderings gives $\pi_k \geq \pi_i$. $\square$

Corollary A.4 (Uniform lower bound). *For every $p \in \Delta_N^\circ$,*

$$H_C(p) \geq \frac{C}{N}.$$

Proof. Using (20),

$$H_C(p) - \frac{C}{N} = \sum_k \left(p_k - \frac{1}{N} \right) \pi_k(p) = \frac{1}{N} \sum_{a < b} (p_a - p_b)(\pi_a(p) - \pi_b(p)).$$

Every summand is nonnegative by Lemma A.3. $\square$

Theorem A.5 (Jacobian master identity). *Let $N \geq 2$, $1 \leq C < N$, and let $q \in \Delta_N^\circ$ be nonuniform. Put $\lambda(\theta) = u + \theta(q - u)$ and $\mathcal{H}(\theta) = H_C(\lambda(\theta))$. Then, for $0 < \theta \leq 1$,*

$$\mathcal{H}'(\theta) = \frac{1}{\theta} \left(H_C(\lambda(\theta)) - \frac{C}{N} \right) + \frac{1}{N\theta} \sum_{1 \leq i < k \leq N} G_{ik}(\lambda(\theta)) (\lambda_i(\theta) - \lambda_k(\theta))^2. \quad (24)$$

In particular $\mathcal{H}'(\theta) > 0$ for $0 < \theta \leq 1$.

Proof. Write $\lambda = \lambda(\theta)$ and $\dot{\lambda}_i = q_i - 1/N$. By the chain rule and (20),

$$\mathcal{H}'(\theta) = \sum_k \dot{\lambda}_k \pi_k + \sum_k \lambda_k \sum_i \frac{\partial \pi_k}{\partial \lambda_i} \dot{\lambda}_i =: T_1 + T_2.$$

Since $\dot{\lambda}_k = (\lambda_k - 1/N)/\theta$, the first term is

$$T_1 = \frac{1}{\theta} \sum_k \left(\lambda_k - \frac{1}{N} \right) \pi_k = \frac{1}{\theta} \left(H_C(\lambda) - \frac{C}{N} \right).$$

For the second term, use Lemma A.1 and (23):

$$\begin{aligned} T_2 &= \sum_i \lambda_i \dot{\lambda}_i \sum_{k \neq i} \lambda_k G_{ik} - \sum_{i \neq k} \lambda_k^2 \dot{\lambda}_i G_{ik} \\ &= \sum_{i \neq k} G_{ik} \lambda_k \dot{\lambda}_i (\lambda_i - \lambda_k). \end{aligned}$$

Symmetrizing over $i \leftrightarrow k$ gives

$$T_2 = \sum_{i < k} G_{ik} (\lambda_i - \lambda_k) (\lambda_k \dot{\lambda}_i - \lambda_i \dot{\lambda}_k).$$

Along the radial path,

$$\lambda_k \dot{\lambda}_i - \lambda_i \dot{\lambda}_k = \frac{1}{\theta} \left[\lambda_k \left(\lambda_i - \frac{1}{N} \right) - \lambda_i \left(\lambda_k - \frac{1}{N} \right) \right] = \frac{\lambda_i - \lambda_k}{N\theta}.$$

Thus

$$T_2 = \frac{1}{N\theta} \sum_{i < k} G_{ik} (\lambda_i - \lambda_k)^2.$$

Adding T_1 gives (24). The first term is nonnegative by Corollary A.4. The second term is strictly positive because $G_{ik} > 0$ for all pairs and $\lambda(\theta)$ is nonuniform when $0 < \theta \leq 1$. Hence $\mathcal{H}'(\theta) > 0$. $\square$

Remark A.6 (Relation to the main kernel). The appendix proof yields a different positive decomposition of the same radial derivative. It splits the derivative into a nonnegative uniform lower-bound term and a strictly positive occupancy-sensitivity square. The main proof instead packages the whole derivative into the explicit subset kernel $K_{ab}^{(C)}$ in (2). The two views are complementary: the subset proof gives the cleanest global certificate, while the Jacobian proof explains the local mechanism.

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